Scientific Method - Probe Experiment

Summary: Define Learning Algorithm (CRITICAL - Blocking) Fix Security Issues (CRITICAL - Blocking) Specify Experiment Design (CRITICAL - Blocking) Validate Integration Points (HIGH -...

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What Happened

> Define Learning Algorithm (CRITICAL - Blocking) Fix Security Issues (CRITICAL - Blocking) Specify Experiment Design (CRITICAL - Blocking) Validate Integration Points (HIGH - Should do before implementation) Simplify Architecture (HIGH - Reduces complexity) Begin Implementation (After CRITICAL issues fixed).

Scientific Method Applied: ✅ Complete Ready for Implementation: ❌ NO (CRITICAL issues must be fixed first) Recommendation: Fix CRITICAL issues, then proceed with implementation using this scientific method framework.

> Date: 2026-01-14 09:57:00 PST Experiment: OriginPoint Probe System Design & Implementation Scientific Method: Full workflow (Hypothesis → Experiment → Analysis).

H4: "Hybrid exploration (random → systematic) outperforms pure random or pure systematic" Prediction: Hybrid exploration finds better patterns faster Verification: Pattern discovery rate is higher with hybrid exploration.

Phases: Phase 1: Security & Core ProbeBeing (Foundation) Phase 2: Learning Algorithm & Scientific Method Integration Phase 3: Feedback Loops & Adaptation Phase 4: Integration & Full System Testing.

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1. Codebase State: Probe system: NOT IMPLEMENTED Being system: ✅ Implemented (v0.5.3) Reality system: ✅ Implemented Scientific method tool: ✅ Implemented D&D character system: ❌ Not implemented.

> 1. Integration State: Reality observation API: ❓ Unclear (needs validation) Scientific method tool compatibility: ❓ Untested Being system integration path: ❓ Unclear.

Phase 4: Integration & Testing (Week 4) ✅ Validate Reality observation API ✅ Test scientific method tool compatibility ✅ Test full Probe cycle (observe → reflect → learn) ✅ Integration testing with Being system.

> 1. Integration State: Reality observation: ✅ Working (or ❌ Failed) Scientific method tool: ✅ Compatible (or ❌ Incompatible) Being system: ✅ Integrated (or ❌ Not integrated).

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1. Scientific Method Report: How scientific method was applied Experiment design evaluation Data collection evaluation Analysis evaluation.

[x] Form hypothesis [x] Design experiment [x] Define variables (independent, dependent, control) [x] Capture initial state (A) [x] Plan data collection (C) [x] Define success criteria [x] Plan analysis methods.

Validate Integration Points: Reality observation API Scientific method tool compatibility Being system integration path.

> Phase 1: Security & Core ProbeBeing (Week 1) ✅ Fix CRITICAL security issues ✅ Implement ProbeBeing class with security ✅ Add ID validation ✅ Set file permissions ✅ Test basic ProbeBeing creation.

Data Validation: All data points must have timestamps All metrics must be numeric (0.0-1.0 or counts) All text data must be sanitized (no sensitive info) All data must be JSON-serializable.

> Statement: "A Probe system that can Observe, Reflect, and Learn using scientific principles will successfully adapt its behavior through feedback loops and hypothesis-driven exploration.".

Prediction: Probe will form testable hypotheses from observations Probe will run experiments to test hypotheses Probe will learn from experiment results and adapt behavior Probe will improve exploration efficiency over time (random → systematic).

Verification Criteria: ✅ Probe forms at least 3 testable hypotheses per 10 observations ✅ Probe runs at least 1 experiment per 5 hypotheses ✅ Probe updates behavior after 70%+ of experiments ✅ Exploration efficiency improves (measured by observation quality / time).

H2: "D&D character stats enhance personality system without conflicts" Prediction: D&D stats add value to personality without overriding existing traits Verification: Personality + D&D stats produces better behavior than personality alone.

> H3: "Collaborative piloting (Probe suggests, AI guides) improves learning rate" Prediction: Collaborative piloting produces faster learning than autonomous or fully controlled Verification: Learning rate (behavior updates / time) is higher with collaborative piloting.

Learning Algorithm Type: Options: Reinforcement Learning, Bayesian Updating, Pattern Matching, Hybrid Default: Hybrid (Reinforcement + Bayesian).

> Exploration Strategy: Options: Pure Random, Pure Systematic, Hybrid (Random → Systematic) Default: Hybrid.

D&D Stat Integration Method: Options: Additive (stats add to personality), Multiplicative (stats multiply), Override (stats replace) Default: Additive.

> Learning Rate: Metric: Behavior updates per experiment Measurement: Count behavior updates / count experiments.

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1. Being System State: Total Beings: 0 (will create ProbeBeing) TheOne Being: ✅ Exists Being storage: `_hidden/.truth/beings/` (exists, permissions: 0700).

> 1. Reality System State: Total Realities: 0 (will create Probe Reality) Reality storage: `_hidden/.truth/realities/` (exists).

Additional Details

1. Scientific Method Tool State: Tool exists: ✅ Components: ✅ All implemented Storage: `scientific_method_tool/experiments/` (exists).

1. Security State: File permissions: Being system uses `0o600` / `0o700` ✅ ID validation: Being system has `_validate_being_id()` ✅ Path validation: Being system validates paths ✅.

Phase 2: Learning Algorithm & Scientific Method (Week 2) ✅ Define learning algorithm ✅ Implement learning algorithm ✅ Integrate scientific method tool ✅ Test hypothesis formation ✅ Test experiment execution.

> Phase 3: Feedback Loops & Adaptation (Week 3) ✅ Implement feedback loops ✅ Implement adaptation mechanism ✅ Test phase transitions (random → systematic) ✅ Test feedback loop effectiveness.

Hypothesis Metrics (Every hypothesis): Timestamp Hypothesis statement Variables (independent, dependent, control) Formation method (pattern recognition, statistical, etc.).

> Experiment Metrics (Every experiment): Timestamp Hypothesis ID Experiment design State A (initial) State B (final) Data collected (C) Result (verified/refuted) Confidence level.

Learning Metrics (Every learning update): Timestamp Trigger (experiment result, feedback, etc.) Update type (belief weight, pattern memory, exploration ratio) Before state After state Success metric.

Adaptation Metrics (Every adaptation): Timestamp Adaptation type (exploration phase, response pattern, etc.) Trigger (success rate, failure rate, etc.) Before state After state.

Collection Frequency: Observations: Real-time (every observation) Hypotheses: Real-time (every hypothesis) Experiments: Per experiment (before/after) Learning: Per update (before/after) Adaptation: Per adaptation (before/after).

> Data Collection Tools: `DataCollector` from `scientific_method_tool/data_collection.py` Custom Probe data collection methods State capture for before/after comparisons.

1. Codebase State: Probe system: ✅ Implemented (or ❌ Failed) Components implemented: Count and list Tests written: Count and coverage Documentation: Count and completeness.

> 1. Probe System State: ProbeBeing instances: Count Experiments run: Count Hypotheses formed: Count Learning updates: Count.

1. Performance State: Observation rate: Observations per minute Hypothesis formation rate: Hypotheses per observation Experiment success rate: Verified / total Learning rate: Updates per experiment.

[ ] Run experiment (implement Probe system) [ ] Collect data (C) during implementation [ ] Monitor progress [ ] Adjust if needed (iterative).

[ ] Capture final state (B) [ ] Analyze results [ ] Verify/refute hypotheses [ ] Generate reports [ ] Document learnings.

> Fix Security Issues: File permissions (`0o600` / `0o700`) ID validation State capture sanitization Reality access control.

Simplify Architecture: Reduce components (8 → 5-6) Combine observation + reflection Simplify feedback loop system.

> Experiment Type: Multi-phase implementation with iterative testing.

Timestamp: 2026-01-14 09:57:00 PST State Type: Initial (Before Probe Implementation).

Timestamp: TBD (After experiment completion) State Type: Final (After Probe Implementation).

State Hash: TBD (calculated from final component states).

> Pattern Recognition: Identify successful patterns Identify failed patterns Extract insights.

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1. Implementation Report: What was implemented What worked What didn't work Lessons learned.

> Define Learning Algorithm: Update mechanism Learning rate Convergence criteria Pattern memory.

Specify Experiment Design: Variables (independent, dependent, control) Measurements Success criteria.

Design Testing Strategy: Unit tests Integration tests Security tests.